

Software Requirement Specification

Project Name: School Reporter

Date: 31 July 2025

Version: 1.0

By: Victor Makhuba

Revision History			
Version	Author	Version Description	Date Completed
1.0	Victor Makhuba	Initial draft of SRS created	

Revision History			
Approving party	Version Approved	Signature	Date

Approval History			
Reviewer	Version Reviewed	Signature	Date

Table of Contents

1. Introduction

- 1.1 Product scope
- 1.2 Product value
- 1.3 Intended audience
- 1.4 Intended use
- 1.5 General description

2. Function requirements

3. External Interface requirements

- 3.1 User interface requirements
- 3.2 Hardware interface requirements
- 3.3 Software interface requirements
- 3.4 Communication interface requirements

4. Non-functional requirements

- 4.1 Security
- 4.2 Capacity
- 4.3 Compatibility
- 4.4 Reliability
- 4.5 Scalability
- 4.6 Maintainability
- 4.7 Usability
- 4.8 Other non-functional requirements

5. Definitions & acronyms

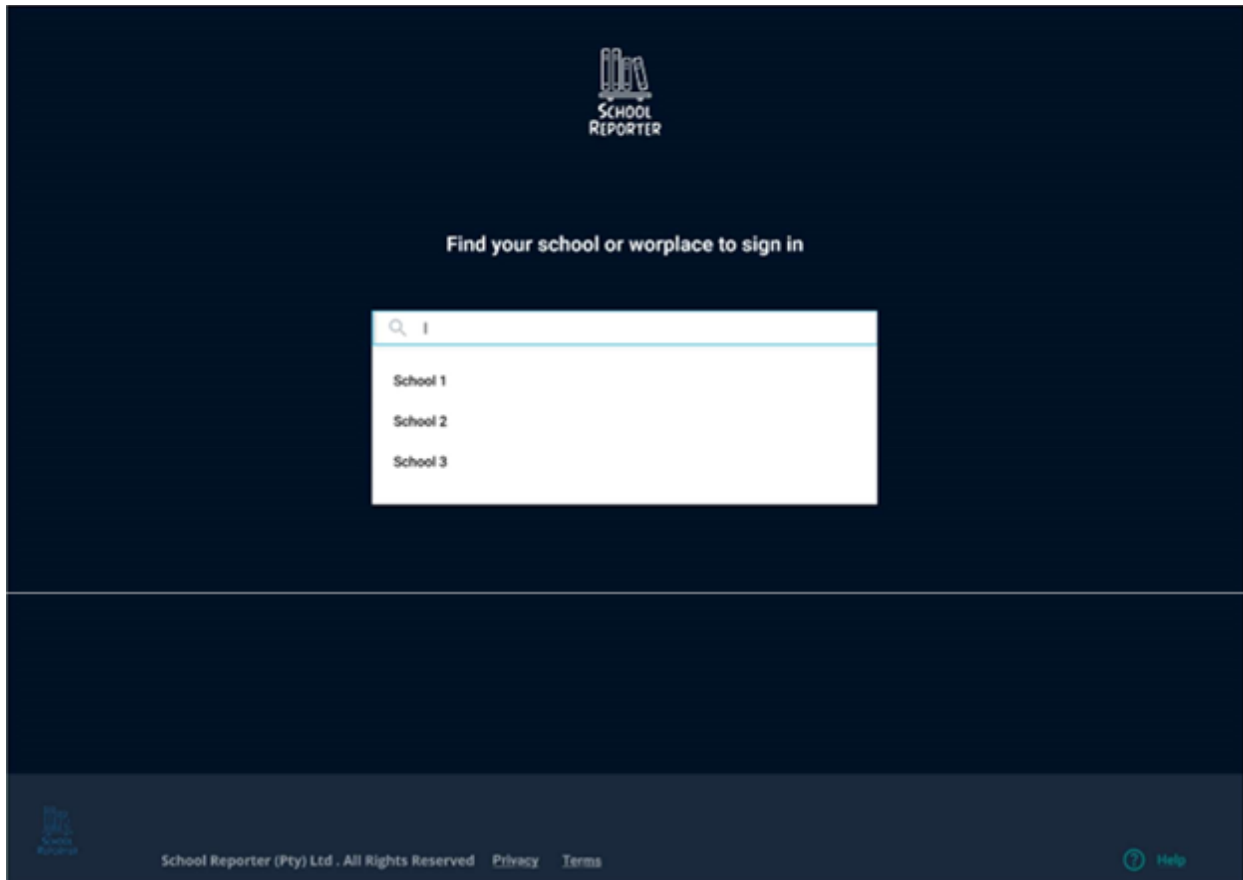
1. Introduction

Describe the purpose of the document. The purpose of this document is to provide a comprehensive specification for the School Reporter System, a web-based platform designed to facilitate communication between schools and parents regarding their children's academic and behavioral performs through WhatsApp or an App	
1.1 Product Scope	<u>List the benefits, objectives and goals of the product.</u> • More parents will be involved in their child's academic life. • It will significantly reduce bunking school or at least make parents aware of it. • More children will complete their homework. • Automate reporting for attendance, grades, and homework.
1.2 Product Value	<u>Describe how the audience will find value in the product.</u> School Reporter provides schools, parents, and learners with a real-time, accessible platform to monitor academic performance, attendance, and communication. It increases transparency, improves engagement, and reduces administrative overhead for educators.
1.3 Intended audience	<u>Write who the product is intended to serve.</u> Schools, learners, parents and teachers.
1.4 Intended use	<u>Describe how the intended audience uses the product.</u> Parents: View attendance, homework, behavior. Teachers: Inform parents about attendance, homework and behavior. Schools/Admins: Manage users, generate reports etc.
1.5 General description	<u>Give a summary of the functions the software would perform, and the features included.</u> School Reporter is a web-based platform with potential mobile integration, offering attendance tracking, homework submission, grade reporting, secure messaging, and fee management. It will integrate with existing school databases where possible.

2. Functional requirements

List of the design requirements, graphics requirements, operating system requirements, and constraints of the product.

Landing page for school selection (Accessed by Teachers)



School Search Functionality

- Users will be presented with a search bar upon initiating the sign-in process.
- As the user types, the system will dynamically display schools that match the entered text, leveraging a type-ahead search feature for efficiency.
- The school list shall be searchable by name, location, or unique identifier assigned to each school within the system's database.

School List Display and Selection

- The system must provide a scrollable list interface that displays schools in alphabetical order when the search bar is activated without input.

- Users should be able to select their school from this list by clicking on the school's name.
- Upon selection, the system will record the choice and direct the user to the appropriate school-specific login page.

School Information Storage

- Each school in the list will have associated metadata stored in the system's database, including, but not limited to, the school's name, address, unique identifier, and any relevant branding assets such as logos.
- The school's metadata must be retrievable and displayed in the subsequent login interface to affirm the user's school selection.

School Data Retrieval

- The system must be capable of retrieving and displaying a list of schools from the database without significant delay, ensuring a responsive user experience.
- Schools data retrieval will implement efficient querying and caching mechanisms to minimize load times and database queries.

Multi-School Support

- The system architecture must support a multi-tenant structure where each school's data is isolated and managed independently while residing within a single application instance.
- Users shall only have access to data pertinent to their selected school, maintaining data integrity and privacy across institutions.

Error Handling and User Feedback

- The system must handle cases where a school cannot be found or if an error occurs during the search process.
- Users should receive clear and concise feedback in the case of search errors or if no results are found, with prompts or instructions on how to proceed.

Accessibility and Compliance

The school selection feature shall be designed to be accessible, following the Web Content Accessibility Guidelines (WCAG) to ensure that all users, including those with disabilities, can navigate and use the feature effectively.

Responsiveness and Device Compatibility

- The school selection interface must be fully responsive, ensuring that it is usable across a variety of devices and screen sizes, from mobile phones to desktop computers.

Login page

[Name] Secondary School

Log into School Reporter

Username
Enter Username

Admin Login

Password
Enter password

[Forgot password?](#)

Remember me ☐

[Sign In](#) →

[Change school or workplace](#)

School Reporter (Pty) Ltd. All Rights Reserved [Terms](#) [Privacy](#) [Help](#)

Account Provisioning

- Each teacher will be provided with a unique username and password upon account creation.
- Usernames should typically be based on a combination of the teacher's name and an identifier from the school to ensure uniqueness across the system.
- Passwords must be automatically generated and meet high-security standards, including a mix of letters, numbers, and special characters. They should be communicated securely to the teachers, ideally through a one-time-use mechanism that requires a password change on the first login.

Authentication Process

- Teachers must log in through a secure login page, entering their provided username and password.
- The system will authenticate teachers by verifying entered credentials against the stored credentials in the database, using hash comparison for passwords.

- Upon successful authentication, the teacher will be granted access to their designated areas of the portal.

Password Security

- Passwords in the system's database must be stored securely, using a strong hashing algorithm with a salt to prevent decryption.
- The system must enforce a minimum password complexity and provide the capability to update passwords periodically to maintain security standards.

Secure Session Management

- After logging in, a secure session must be established using tokens such as JSON Web Tokens (JWTs) to maintain the authentication state.
- Sessions must timeout after a period of inactivity, requiring re-authentication to regain access.

Account Recovery

The portal shall provide a secure method for teachers to recover lost or forgotten usernames and passwords, involving email verification or similar methods to confirm identity before allowing password reset or retrieval.

Multi-Factor Authentication (MFA)

As an additional security layer, the system should support multi-factor authentication. This can be implemented as an optional security feature for teachers to activate.

Authentication Audit Logs

The system should maintain an audit log of all authentication attempts, successful and unsuccessful, to monitor for unauthorized access attempts or other security concerns.

Compliance with Privacy Regulations

The authentication system must comply with relevant privacy and data protection regulations, ensuring that teacher and student data is handled according to legal requirements.

Attendance

2.4 Functional Requirements

Attendance Management

SCHOOL REPORTER

Show 10 entries Search...

Home
Attendance
Homework
Behavior
Announcements
Messages 4

Mrs. Maluleke

Learner ID	Name	Surname	Grade	Attendance	Action
#1	Learner 1	Surname	8	Present	
#2	Learner 2	Surname	8	Present	
#3	Learner 3	Surname	8	Present	
#4	Learner 4	Surname	8	Present	
#5	Learner 5	Surname	8	Present	
#6	Learner 6	Surname	8	Present	
#7	Learner 7	Surname	8	Present	
#8	Learner 8	Surname	8	Absent	
#9	Learner 9	Surname	8	Present	
#10	Learner 10	Surname	8	Select Value Present Absent	

Previous Next Submit

School Reporter (Pty) Ltd. All Rights Reserved Terms Privacy Help

Mrs. Maluleke Sign Out

A sign out drop down will pop out when the user clicks on their name

Add a scroll bar that will accommodate the entire list on the

2.4.1 Class Roster Display

- ▢ The attendance system will display the class roster for each teacher, reflecting the class as per the school's official database.
- ▢ The roster will list students' names and associated information such as grade and student ID.

Class Register Display

- The attendance system will display the class register for each teacher, reflecting the class as per the school's official database (SA-Sams).
- The register will have learners' names and associated information such as grade etc.

Daily Attendance Input

- Teachers can mark each student as present or absent using a dropdown menu for each student entry on the roster.

Attendance Data Reset

- The attendance marks (present/absent) for each student will reset to their default state at the end of each day to ensure that new data can be recorded the following day.
- The reset functionality will not affect the student roster, which remains constant.

Attendance Record Finalization

- Teachers will have the ability to review and finalize attendance records for the day, with a function to submit the final attendance data.

Administrative Editing of Roster

- Admins will have the authority to edit the student roster. This includes adding new learners' or removing learners' who have left the school.
- Edits to the roster by the admin can be performed at any time.

CSV Import for Roster Updates

- The system will allow admins to update the student roster by uploading a CSV file directly to the system.
- Upon CSV upload, admins can select which columns and rows to import, allowing for seamless updates to the class rosters.

Integration with SQL Database

- Attendance data submitted by teachers will automatically be pushed to a designated database for record-keeping and further use by the chatbot system or App.
- This integration must support real-time updates to ensure immediate reflection of attendance data in the chatbot's knowledge base and for the App, the update will happen around the time schools are dismissed with the entire summary of events of the day (Attendance, homework, behavior, announcements).

Intra-Interface Table Scrolling

- The attendance interface will allow teachers to scroll through the list of students within the frame of the table without affecting the rest of the page's UI, ensuring a fixed header for context.
- The interface will be designed to support various device sizes while maintaining usability and readability.

Attendance Reporting and Analytics

- The system should provide attendance reporting functionality for admins and teachers, enabling the generation of attendance reports over selected periods.
- Attendance analytics could include visualization of attendance trends, identification of absenteeism patterns, and more.

Homework management page

SCHOOL REPORTER

Show 10 entries

Home

Attendance

Homework

Behavior

Announcements

Messages 4

Mrs. Maluleke

Learner ID	Name	Surname	Grade	Behavior
#1	Learner 1	Surname	8	
#2	Learner 1	Surname	8	
#3	Learner 1	Surname	8	
#4	Learner 1	Surname	8	
#5	Learner 1	Surname	8	
#6	Learner 1	Surname	8	
#7	Learner 1	Surname	8	
#8	Learner 1	Surname	8	
#9	Learner 1	Surname	8	
#10	Learner 1	Surname	8	

Previous 1 2 3 Next

School Reporter (Pty) Ltd . All Rights Reserved [Terms](#) [Privacy](#)

Help

Homework Roster Display

- The homework management system will display a roster that includes period, class, subject, and assigned homework.
- This roster will mirror the school's class schedule and subject offerings as per the SASAMS database.

Homework Assignment Input

- Teachers can assign homework to their classes using a user-friendly interface.
- For each class period, teachers will select from dropdown menus in the homework column, populated with subjects as outlined by the SA-SAMS.

Administrative Roster Management

- Administrators will have exclusive rights to edit the homework roster, including period, class, and subjects.
- They will be able to add or remove periods, change class assignments, and update subject lists.

Subject List Synchronization with SA-SAMS

- The system will synchronize the available subjects for homework assignment with those listed in the SA-SAMS, ensuring all subjects offered by the school are selectable.
- This synchronization should be maintained regularly to reflect any changes in the curriculum offerings.

CSV Import for Roster Updates

- Similar to attendance management, the system will allow administrators to update the homework roster by uploading a CSV file.
- Admins can select specific data to import, making updates to the roster efficient and accurate.

Real-time Data Capture and Export

- Homework data entered by teachers will be captured in real-time, with the ability to export data for reporting purposes.
- The export feature should support various formats and allow administrators to generate reports for individual classes, grades, or subjects.

Access Control and Permissions

- The system must implement role-based access controls to ensure that only administrators can edit the roster while teachers can assign and track homework.

Integration with Communication Channels

- The homework management system should be capable of integrating with the school's communication channels to notify students and parents of new assignments, due dates, and submission reminders.

Behavior Report page

SCHOOL REPORTER

Show 10 entries

Home

Attendance

Homework

Behavior

Announcements

Messages 4

Mrs. Maluleke

Learner ID	Name	Surname	Grade	Behavior
#1	Learner 1	Surname	8	
#2	Learner 1	Surname	8	
#3	Learner 1	Surname	8	
#4	Learner 1	Surname	8	
#5	Learner 1	Surname	8	
#6	Learner 1	Surname	8	
#7	Learner 1	Surname	8	
#8	Learner 1	Surname	8	
#9	Learner 1	Surname	8	
#10	Learner 1	Surname	8	

Previous 1 2 3 Next

School Reporter (Pty) Ltd . All Rights Reserved [Terms](#) [Privacy](#) [Help](#)

Behavior Incident Logging

- Teachers can log behavior incidents for each student using the system.
- The interface for logging behavior will have an Edit icon under behavior in the table where teachers can click and it will send them to a page where they can w

Behavior Category Selection

- For each student, teachers will have a dropdown menu in the behavior column, populated with predefined behavior categories.
- These categories will include but are not limited to positive behaviors, disruptions, absenteeism, and disciplinary actions, aligned with the school's behavior policy.

Daily Behavior Record

- Teachers can update the behavior status for each student daily or as incidents occur.

- The system will save behavior records at the end of each day and be prepared for a new entry the next school day.

Behavior Data Reset and Finalization

- Similar to attendance, behavior entries will have a time window within which teachers can edit them before they are finalized for the day.
- Once finalized, the records are locked to prevent further changes, ensuring data integrity.

Administrator Control Over Behavior Categories

- Administrators can edit the behavior categories in the system to update or modify the list according to school policy changes.
- Admins will also have the ability to handle exceptional behavior records that require attention beyond the teacher's scope.

Student Behavior History

- The system will maintain a history of each student's behavior records, allowing for trend analysis and individual student reports.
- Behavior trends can be monitored over time to identify patterns and areas for intervention.

Behavior Reporting

- Administrators and teachers can generate reports on student behavior, class-wide behavior patterns, or incidents over selected periods.
- Reports can be exported and used for parent-teacher meetings, counseling sessions, or as part of student profiles.

Integration with Google Sheets and Chatbot

- Behavior records entered by teachers will be synchronized with the designated Google Sheet, as with attendance and homework.
- This integration will provide the chatbot system with the latest behavior data for communication with parents.

- The system shall enforce strict confidentiality measures to protect the integrity and privacy of behavior records.
- Access to behavior data shall be governed by a role-based access control system, ensuring users can only access information pertinent to their role within the school.
- Teachers will be able to access and input data only for their assigned students to maintain data compartmentalization.
- Administrators will have overarching access to all behavior data for oversight and management purposes, with the ability to grant or revoke access permissions as necessary.
- Audit logs shall be maintained to track all access and changes made to the behavior records, providing an immutable record of all actions for accountability.
- All data transmission involving behavior records shall be encrypted using secure protocols to prevent interception or unauthorized access during transit.
- Behavior data will be stored in compliance with applicable data protection laws, and the system will include mechanisms for data anonymization and secure deletion when required.
- Regular security audits and penetration testing shall be conducted to ensure the access control mechanisms are robust and safeguard against unauthorized access.

External interface requirements

3.1 User interface requirements	Describe the logic behind the interaction between the users and the software (screen layout, style guides, etc.)
3.2 Hardware interface requirements	List the supported devices the software is intended to run on, the network requirements, and the communication protocols to be used
3.3 Software Interface requirements	Include the connection between your software and other software components, including frontend/backend frameworks, libraries etc.
3.4 Communication interface requirements	List any requirements for communication programs your product will use, ike emails or embedded forms.

3. Non-functional requirements

4.1 Security	Include any privacy and data protection that should be adhered to
4.2 Capacity	Describe the current and future storage needs for your software
4.3 Compatibility	List the minimum hardware requirements for your software
4.4 Reliability	Calculate what the critical failure time of your product would be under normal usage
4.5 Maintainability	Describe how continuous integration should be used to deploy features bug fixes quickly
4.7 Usability	Describe how easy it should be for end-users to use your software
4.8 Other	List any additional non-functional requirements

Definition and Acronyms